

Nikhil Sharma

Data Engineer & ML Engineer

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Experience

Trainee Data / ML Engineer

Sep 2025 – Present

Key Management Technologies (KMG)

- Built and optimized PySpark + Databricks pipelines on Medallion architecture processing 100K+ records/day, reducing latency by 25%.
- Developed backend services and ETL workflows using FastAPI across PostgreSQL, MySQL, and SQL Server environments.
- Built a LangChain + LangGraph workflow with modular nodes for extraction, classification, field extraction, and storage, enabling automated processing of heterogeneous input documents into structured outputs.

Projects

NL-to-SQL Platform (RAG-Based)

Dec 2025 – Mar 2026

- Built production-grade RAG-based NL-to-SQL system enabling natural language querying for non-technical users.
- Designed hybrid retrieval pipeline (vector + BM25 + reranking), improving query accuracy and reducing hallucinated SQL.
- Developed modular FastAPI backend with ingestion pipeline (schema → chunking → embedding → indexing) and secure SQL validation.

Heart Disease Risk Prediction

Jun 2023

- Developed ML-based system for early cardiovascular risk prediction using clinical datasets.
- Compared Logistic Regression, Random Forest, and Neural Networks, improving accuracy by 30%.
- Built automated prediction pipeline reducing diagnosis time by 40%.

Publications

The Sound of Sentiment: Optimizing Emotion Recognition in Speech with Neural Networks Dec 2024
ICSCNA 2024 · IEEE Xplore

- Built deep learning pipeline using MFCC and Mel spectrogram features on RAVDESS dataset (1400 samples).
- Evaluated CNN, LSTM, GRU models, achieving 85% accuracy and F1-score up to 0.93.
- Improved model generalization using preprocessing, normalization, and feature engineering techniques.

Skills

Languages: Python, C#

Data Engineering: PySpark, Databricks, ETL/ELT, Data Modeling

Machine Learning TensorFlow, Scikit-learn, CNN, LSTM, GRU

GenAI: Rag PipeLines, Hybrid Retrieval, LangChain, LangGraph

Backend: FastAPI, Flask, REST APIs

Databases: PostgreSQL, MySQL, SQL Server

Tools: Git, GitHub, VS Code

Education

B.Tech — Computer Science, Chandigarh University

2021 – 2025

GPA: 7.69 / 10 — 100% Merit Scholarship

Senior Secondary (XII), Vidya Sagar Sr. Sec. School, Jaipur

2021

78.2%

Certifications

- Advanced Machine Learning on Google Cloud 2024
- Introduction to Data Science 2023